

PULAK SEN

Full stack developer

+91 8116747805 | pulaksen.dev@gmail.com | linkedin.com/in/pulak-sn/ | github | Portfolio | Siliguri, India

SKILLS

Programming: JavaScript, TypeScript, Python, C, Java, SQL

Frontend: React.js, Next.js, Tailwind CSS, ShadCN UI

Backend: Node.js, Express.js, Nest.js, REST APIs, LangChain, JWT Authentication, Prisma

Database: MySQL, PostgreSQL, MongoDB

DevOps & Tools: Kafka, Redis, Ollama (Ollama 3.2), RabbitMQ, Docker, Kubernetes, Git, GitHub

PROJECTS

Event-Flow | Microservices-based Event Booking System [\[Source code\]](#)

- Engineered a Nest.js monorepo for a ticket booking system with five **microservices**—API Gateway, Auth, Events, Tickets, and Notifications—utilizing Kafka for scalable, asynchronous inter-service communication.
- A notification engine that consumes **Kafka topics** to trigger automated user alerts and a centralized API Gateway to streamline routing across the ticket booking ecosystem.
- Tech stack: TypeScript, Nest.js, Kafka, KafkaJS, PostgreSQL, MailHog, Docker

PDF-RAG Chat Application [\[Source code\]](#)

- Developed an AI-powered document intelligence platform that enables users to upload PDF files and interact with their content through natural language conversations using semantic search and RAG.
- Engineered an asynchronous document processing pipeline with **BullMQ** and **Valkey** to extract, chunk, embed, and index PDF content in **Qdrant**, integrating **LangChain** and **Ollama** to generate context-aware responses.
- Tech stack: JavaScript, Node.js, Express.js, BullMQ, LangChain, Ollama, Qdrant, Valkey, Multer, Docker

Backend Ledger [\[Source code\]](#)

- A financial transaction system using OAuth 2.0, Nodemailer, and a **dual-table (Transaction & Ledger)** architecture, leveraging idempotency keys and **MongoDB aggregation pipelines** to ensure data consistency, secure notifications, accurate debit/credit reconciliation, auditability, and real-time balance computation.
- Tech stack: JavaScript, Node.js, Express, MongoDB, Mongoose, OAuth 2.0, Nodemailer, JWT, Cookie-parser.

Real-Time Messaging Platform [\[Source code\]](#)

[\[Live Demo\]](#)

- Built a full-stack real-time messaging platform using the **MERN stack** and **Socket.IO**, implementing JWT-based authentication, secure REST APIs and WebSocket connections, private messaging, online presence tracking, notification alerts, and image sharing via **Cloudinary**
- Tech stack: JavaScript, React, Node.js, Express, MongoDB, Socket.IO, Cloudinary, DaisyUI, Tailwind CSS.

EDUCATION

South Asian University, New Delhi , M.Sc. in Computer Science

August 2021 – June 2023

- GPA: 5.85/9.0
- Coursework:** Computer Organization and Design, Mathematical Foundations of Computer Science, Blockchain Technology, Cryptography and Network Security, Theory of Computation, Design and Analysis of Algorithms, IoT

A.P.C Roy Govt. College, Siliguri , B.Sc. in Computer Science

July 2018 – August 2021

- GPA: 8.57/10
- Coursework:** Discrete Structures, Software Engineering, Programming Fundamentals using C, Programming in JAVA, Data Structure, Operating System, Networking, DBMS, Computer System Architecture, Numerical Method

ACHIEVEMENTS

- President's Scholarship:** Awarded for M.Sc. at South Asian University for securing the **1st position (Rank 1)** among all candidates in the national-level entrance examination. [\[View Certificate\]](#)